Experiment No. 2

Experiment Name

Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

part-1 Python code

```
#step-1
import cv2
import numpy as np
import matplotlib.pyplot as plt
#step-2
img = cv2.imread("image.jpg")
# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
plt.imshow(gray, cmap='gray')
plt.title("Original Image")
plt.axis("off")
plt.show()
#step-3
plt.hist(gray.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()
#step-4
min_val = np.min(gray)
max_val = np.max(gray)
stretched = ((gray - min_val) / (max_val - min_val) * 255).astype(np.uint8)
plt.imshow(stretched, cmap='gray')
plt.title("Contrast Stretched Image")
plt.axis("off")
plt.show()
#step-5
equalized = cv2.equalizeHist(gray)
plt.imshow(equalized, cmap='gray')
plt.title("Histogram Equalized Image")
plt.axis("off")
plt.show()
#step-6
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_img = clahe.apply(gray)

plt.imshow(clahe_img, cmap='gray')
plt.title("CLAHE Enhanced Image")
plt.axis("off")
plt.show()
# step-7
plt.figure(figsize=(10, 6))
plt.hist(gray.ravel(), 256, [0, 256], alpha=0.5, label="Original")
plt.hist(equalized.ravel(), 256, [0, 256], alpha=0.5, label="Equalized")
plt.hist(clahe_img.ravel(), 256, [0, 256], alpha=0.5, label="CLAHE")
plt.title("Histogram Comparison")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.legend()
plt.show()
#step-8
plt.figure(figsize=(12, 8))
plt.subplot(2, 2, 1)
plt.imshow(gray, cmap='gray')
plt.title("Original")
plt.axis("off")
plt.subplot(2, 2, 2)
plt.imshow(stretched, cmap='gray')
plt.title("Contrast Stretching")
plt.axis("off")
plt.subplot(2, 2, 3)
plt.imshow(equalized, cmap='gray')
plt.title("Histogram Equalization")
plt.axis("off")
plt.subplot(2, 2, 4)
plt.imshow(clahe_img, cmap='gray')
plt.title("CLAHE")
```

```
plt.axis('off')
plt.tight_layout()
plt.show()
#step-9
pip install opencv-python numpy matplotlib
```

Part - 2 Question

1. What is image contrast, and why is it important in image processing?

Image contrast is the difference between the bright and dark regions of an image. High contrast makes objects and edges easier to see, while low contrast makes an image appear dull. It is important for better visibility, feature extraction, segmentation, and object recognition.

2. Explain the concept of an image histogram. What information does it provide?

An image histogram is a graph showing the frequency of different pixel intensity values in an image.

- X-axis ⟶ Pixel intensity (0–255 for an 8-bit image)
- Y-axis ⟶ Number of pixels It tells us whether an image is dark, bright, or has good contrast.

3. Differentiate between Histogram Stretching and Histogram Equalization.

Histogram Stretching	Histogram Equalization
Expands the existing intensity range.	Redistributes intensity values.
Improves dynamic range.	Improves overall contrast.
Usually uses minimum and maximum intensity.	Uses the cumulative distribution function (CDF).
Simple technique.	More powerful for uneven intensity distributions.

4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

CLAHE (Contrast Limited Adaptive Histogram Equalization) improves contrast by dividing an image into small regions and performing histogram equalization locally.

Difference from standard HE Standard HE works on the whole image. CLAHE works on small regions (tiles). CLAHE limits contrast amplification, reducing noise amplification. It is useful for images with varying illumination.

5. Why is histogram equalization commonly applied before feature extraction and image segmentation? It improves image contrast and makes important features such as edges, boundaries, and textures more visible. This helps feature extraction and segmentation algorithms work more effectively.

6. Mention three real-world applications where histogram equalization is widely used.

Medical imaging – X-rays, CT scans, MRI images.

Satellite/remote sensing images.

Low-light photographs and surveillance images.

These applications are specifically included in the experiment's evaluation examples.

7. What are the limitations of global histogram equalization?

Can over-enhance the image.

May amplify noise.

Can lose some brightness information.

Does not handle local illumination variations well.

May produce unnatural-looking images.

8. How does contrast enhancement improve the performance of object detection and recognition systems?

Contrast enhancement makes objects, edges, and important features more distinct from the background. This helps computer vision systems detect and recognize objects more accurately.

9. Compare histogram-based enhancement techniques with brightness adjustment methods.

Histogram-based enhancement:

Changes the distribution of pixel intensities.

Improves contrast.

Can reveal hidden details.

Brightness adjustment:

Mainly shifts pixel intensities brighter or darker.

Does not necessarily improve contrast.

May cause loss of details if over-applied.

10. Why is CLAHE preferred for medical imaging and low-light image enhancement?

CLAHE is preferred because it improves local contrast while limiting excessive contrast enhancement. It works well when different parts of an image have different illumination and helps reveal details without strongly amplifying noise.